The Superior University, Lahore

Faculty of Economics & Commerce

Course Module

Course Information

Course Title

Applications of Information and Communication Technologies – LAB

Course ID

ICT-Lab-101

Course Type

Computing

Credit hours

1

Hours per week (C-L)

1

Date

Preferred Semester

1

Offered Course Information

Programs

BS in Fintech

Semester Session

1

Instructor(s)

Mr. Junaid Sabri

TA / Junior Lecturer

Course Description

This course provides hands-on, practical training in core business computing tools used across the FinTech industry. Students will develop structured document design skills using MS Word, create professional presentations using MS PowerPoint, and apply data management and analysis techniques using MS Excel—including logical functions (IF, Nested IF) and data retrieval tools (VLOOKUP). The course concludes with an integrative project that connects all three tools in a real-world business context. n line with the Islamic values of Amanah (trust) and precision, this lab emphasizes using data analytics and software tools with absolute integrity, ensuring fair documentation and preventing digital plagiarism

Program Educational Objectives (PEO)

PEO No.

Program Educational Objective (PEO)

3C Pillar Alignment

PEO 1

FinTech Expertise: Apply integrated knowledge of finance, computer science, data analytics and emerging technologies to design, build and evaluate digital-finance solutions.

Competency

PEO 2

Innovation & Entrepreneurship: Launch, scale, or transform products, services, and ventures that create value in the digital-finance ecosystem.

Courage

PEO 3

Ethical, Secure & Compliant Practice: Uphold ethical standards, cyber security best practices, and regulatory compliance while protecting consumers and markets.

Character

PEO 4

Global & Sustainable Impact: Address global trends and societal needs, financial inclusion, green finance, Shariah compliance, through responsible FinTech.

Character

Courage

Program Learning Outcomes (PLO)

PLO No.

Program Educational Objective (PEO)

3C Pillar Alignment

PLO 2

Skillful Demonstrate an understanding of statistical, machine learning, and AI tools to extract meaningful insights from complex financial data and support informed financial decision-making and organizational performance.

Competence

Courage

PLO 9

Communication An ability to communicate complex technical and financial ideas clearly and effectively, orally as well as in writing, with diverse teams and the broader business and technology community, such as being able to comprehend and produce clear technical reports and financial documentation, make effective presentations, and give and receive precise and informed instructions in multidisciplinary and collaborative settings.

Character

Course Objectives (CO)

The objective of this course is to enable students to understand;

No.

Outcome

BT Level

Relation with PLO

Relation with 3C

CLO1

Construct professional business documents by applying advanced word processing, structural layouts, and formatting tools.

P3

PLO 2

Competence

CLO2

Develop impactful business presentations and apply logical and analytical tools

for spreadsheet-based data management.

P-4

PLO 2,9

Competence

Character

Courage

Lecture type

This course employs an Inquiry-Based Learning method, using multimedia-guided demonstrations and live workstation practice.

Prerequisites

-

Follow up

Courses

Introduction to Database - DB-10

Course Software

or Tool

MS Word, MS PowerPoint, MS Excel

Textbook

Title

Edition

Authors

Publisher

Year

Recommend Book

Discovering Computers & Microsoft Office 365, Office 2016

1st

Steven M. Freund

Mary Z. Last

CENGAGE Learning

2017

References

Assessment

Criteria (100%)

Assessment

Weight

CLO

Mode

BTL

3C Alignment

Lab Tasks

40%

CLO 1

Soft Form – Submission on LMS

P3

Competence

Project

40%

CLO 2

Soft Form – Submission on LMS

P3

Competence

Viva

20%

CLO 2

Viva

P4

Competence

Character

Methods of

Evaluation

Lab Tasks, Project and Viva

Week #

Topic

Learning Objective

Taxonomy

MIT

Assessment Tool

CLO

3C

PLO

W1-2

Hardware Identification

Identify CPU, RAM, and I/O devices physically and logically.

P3

Demo & Practice

Identification Quiz

CLO 1

Comp.

PLO 2

W3

Word: Basic Formatting

Apply font styles and professional formatting to text.

P3

Live Demo

Formatting Task

CLO 1

Comp.

PLO 9

W4

Paragraphs & Visuals

Construct workflows using SmartArt and paragraph tools.

P3

Practical Drill

Workflow Design

CLO 1

Comp.

PLO 9

W5

Page Layout & Identity

Configure page setup, headers, and document security.

P3

Hands-on Lab

Layout Assignment

CLO 1

Char.

Cour.

PLO 9

W6

Tables & TOC

Design complex tables and automated Table of Contents.

P3

Step-by-Step

Document Structure

CLO 1

Comp.

PLO 9

W7

AI & Modern Tools

Utilize AI prompts and Canva for digital content drafting.

P3

Inquiry-Based

AI Content Draft

CLO 1

Comp.

PLO 2

W8

Mid-Term Exam

Practical assessment of Word & Hardware tasks.

P3

Practical Test

Lab Exam

CLO 1

Comp.

Cour.

PLO 2,9

W9-10

Presentation Slides

Develop interactive slides with animations and transitions.

P3

Creative Demo

Slide Deck Project

CLO 2

Char.

PLO 9

W11

Excel: Data Basics

Execute data entry and basic arithmetic (Sum, Max, Min).

P2

Practical Lab

Basic Workbook

CLO 2

Comp.

PLO 2

W12

Excel: Logic (IF)

Apply IF/Nested IF functions for automated grading.

P3

Problem Solving

Result Sheet Task

CLO 2

Comp.

PLO 2

W13

Excel: Lookup Tools

Implement VLOOKUP for automated business databases.

P3

Live Case Study

Inventory Task

CLO 2

Comp.

PLO 9

W14

Integration: Mail Merge

Link Excel data with Word for automated mail merge.

P4

Tool Integration

Mail Merge Task

CLO 2

Comp.

PLO 2,9

W15

Analytical Tools

Summarize data via Pivot Tables and graphical charts.

P4

Data Analytics

Analysis Report

CLO 2

Comp.

PLO 2

W16

Final Project

Integrated project covering Word, PPT, and Excel.

P4

Presentation

Final Viva & Project

CLO 2

Char.

PLO 2,9

1. Lab Tasks (40% of Total Grade)

Evaluates weekly/regular hands-on technical proficiency, tool utilization, and execution.

Criteria

Excellent (85-100%)

Good (70-84%)

Developing (50-69%)

Poor (Below 50%)

Technical Execution

All tasks completed with zero errors; outputs match specified requirements perfectly.

Tasks completed with minor errors or formatting issues that don't affect functionality.

Partial completion; major functional bugs or missing key requirements.

Tasks are incomplete, non-functional, or not submitted.

Tool & Software Mastery

Demonstrates advanced, efficient use of software/tools (e.g., shortcuts, correct syntax, clean layouts).

Uses software appropriately to achieve the outcome, though methods could be optimized.

Struggles with tool navigation; relies heavily on basic, inefficient methods.

Inability to use the required software or tools independently.

Problem Solving & Logic

Independent troubleshooting; handles edge cases smoothly without prompting.

Solves core problems but needs guidance or hints for unexpected errors.

Requires constant step-by-step assistance to resolve basic errors.

Shows no evidence of independent problem-solving effort.

2. Project (40% of Total Grade)

Evaluates long-term synthesis, application of ICT concepts to a real-world scenario, and documentation.

Criteria

Excellent (85-100%)

Good (70-84%)

Developing (50-69%)

Poor (Below 50%)

Scope & Complexity

Project addresses a complex, real-world problem with an innovative, comprehensive solution.

Project is well-defined and complete, but sticks entirely to safe, basic concepts.

Project is overly simplistic or lacks a clear, practical purpose.

Fragmented concept; fails to address the minimum project requirements.

Implementation & Quality

Seamless integration of ICT tools; robust, user-friendly, and completely bug-free.

Fully functional with minor UI/UX or data handling quirks.

Partially functional; core features work but system is unstable or prone to breaking.

Non-functional project; key components fail to integrate or operate.

Documentation & Design

Exceptional documentation (clear structure, architecture/flow charts, clean formatting).

Good documentation covering all sections, though some explanations are brief.

Poorly organized documentation; lacks clear instructions or design diagrams.

Missing or copied documentation; minimal effort shown.

3. Viva / Oral Defense (20% of Total Grade)

Evaluates individual understanding, conceptual depth, and communication skills to ensure authorship.

Criteria

Excellent (85-100%)

Good (70-84%)

Developing (50-69%)

Poor (Below 50%)

Conceptual Clarity

Explains why specific tools or methodologies were used with deep theoretical backing.

Understands what was done, but struggles slightly to link it to broader ICT theories.

Memorized steps but cannot explain the underlying principles or logic.

Completely unable to explain concepts or how things work.

Code/Project Familiarity

Can instantly locate, explain, and modify any part of their lab work or project code/files.

Knows the project well but takes time to locate specific logic or configurations.

Hesitant; struggles to explain chunks of their own project, suggesting heavy reliance on others.

Unfamiliar with the project structure; clear signs of plagiarism or ghost-authoring.

Communication Skills

Answers are concise, technically accurate, confident, and professionally delivered.

Clear answers, though uses some non-technical language or needs questions rephrased.

Answers are vague, highly hesitant, or heavily rely on guessing.

Unable to articul